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United States Patent	12415364
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Miyashita; Takeho et al.

Liquid ejection head and liquid ejection apparatus

Abstract

A liquid ejection head and liquid ejection apparatus, which are capable of preventing deterioration in ink circulation efficiency in the vicinities of ejection ports, includes a common supply channel and a common collection channel as separate channels. Ink supplied from the common supply channel is supplied to pressure chambers through supply connection channels, and collected from the pressure chambers into the common collection channel through collection connection channels. Also, the common supply channel and the common collection channel extend in a direction crossing a main scanning direction in which the liquid ejection head is scanned and the ejection direction of the liquid.

Inventors: Miyashita; Takeho (Kanagawa, JP), Muraoka; Chiaki (Saitama, JP), Toda; Kyosuke (Kanagawa, JP), Yoshikawa; Shimpei (Kanagawa, JP)

Applicant: CANON KABUSHIKI KAISHA (Tokyo, JP)

Family ID: 1000008819887

Assignee: Canon Kabushiki Kaisha (Tokyo, JP)

Appl. No.: 18/082918

Filed: December 16, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230120077 A1	Apr. 20, 2023

Foreign Application Priority Data

JP	2021-205258	Dec. 17, 2021
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Publication Classification

Int. Cl.: B41J2/14 (20060101); B41J2/175 (20060101); B41J2/18 (20060101); B41J25/00 (20060101)

U.S. Cl.:

CPC B41J2/18 (20130101); B41J2/1404 (20130101); B41J25/001 (20130101); B41J2/17556 (20130101); B41J2202/12 (20130101)

Field of Classification Search

CPC: B41J (2/18); B41J (2/1404); B41J (25/001); B41J (2/17556); B41J (2202/12)

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Primary Examiner: Ameh; Yaovi M

Attorney, Agent or Firm: Venable LLP

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a liquid ejection head and a liquid ejection apparatus including

the liquid ejection head.

Description of the Related Art

(2) Japanese Patent Laid-Open No. 2011-098491 discloses a liquid ejection apparatus that ejects an ink from an inkjet head. This liquid ejection apparatus is capable of circulating the ink.

(3) However, in a configuration as disclosed in Japanese Patent Laid-Open No. 2011-098491, the ink circulation efficiency is low at the nozzles. Hence, the ink in the nozzles tends to get thickened.

SUMMARY OF THE INVENTION

(4) Thus, the present invention provides a liquid ejection head and liquid ejection apparatus capable of preventing a deterioration in ink circulation efficiency in the vicinities of ejection ports.

(5) A liquid ejection head of the present invention includes: an ejection module configured to eject a liquid in an ejection direction, and having a plurality of ejection ports, a plurality of energy generation elements configured to eject the liquid respectively from the plurality of ejection ports, a plurality of pressure chambers communicating respectively with the plurality of ejection ports, a plurality of supply channels configured such that the liquid is supplied therethrough respectively to the plurality of pressure chambers, and a plurality of collection channels configured such that the liquid is collected therethrough respectively from the plurality of pressure chambers; and a circulation unit configured to circulate the liquid by supplying the liquid to the plurality of supply channels of the ejection module and collecting the liquid from the plurality of collection channels of the ejection module, in which the liquid ejection head is capable of moving in a main scanning direction, and the supply channels and the collection channels extend in a direction crossing the main scanning direction and the ejection direction.

(6) According to the present invention, it is possible to provide a liquid ejection head and liquid ejection apparatus capable of preventing a deterioration in ink circulation efficiency in the vicinities of ejection ports.

(7) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a view describing a liquid ejection apparatus;

(2) FIG. 1B is a diagram describing the liquid ejection apparatus;

(3) FIG. 2 is an exploded perspective view of a liquid ejection head;

(4) FIG. 3A is a vertical cross-sectional view of the liquid ejection head;

(5) FIG. 3B is an enlarged cross-sectional view of an ejection module;

(6) FIG. 4 is a schematic external view of a circulation unit;

(7) FIG. 5 is a vertical cross-sectional view illustrating a circulation path;

(8) FIG. 6 is a block diagram schematically illustrating the circulation path;

(9) FIG. 7A is a cross-sectional view illustrating an example of pressure adjustment units;

(10) FIG. 7B is a cross-sectional view illustrating the example of the pressure adjustment units;

(11) FIG. 7C is a cross-sectional view illustrating the example of the pressure adjustment units;

(12) FIG. 8A is a perspective external view of a circulation pump;

(13) FIG. 8B is a perspective external view of the circulation pump;

(14) FIG. 9 is a cross-sectional view of the circulation pump illustrated in FIG. 8A along the IX-IX line;

(15) FIG. 10A is a diagram describing a flow of an ink inside the liquid ejection head;

(16) FIG. 10B is a diagram describing the flow of the ink inside the liquid ejection head;

(17) FIG. 10C is a diagram describing the flow of the ink inside the liquid ejection head;

(18) FIG. 10D is a diagram describing the flow of the ink inside the liquid ejection head;

- (19) FIG. 10E is a diagram describing the flow of the ink inside the liquid ejection head;
- (20) FIG. 11A is a schematic view illustrating a circulation path in an ejection unit;
- (21) FIG. 11B is a schematic view illustrating the circulation path in the ejection unit;
- (22) FIG. 12 is a view illustrating an opening plate;
- (23) FIG. 13 is a view illustrating an ejection element substrate;
- (24) FIG. 14A is a cross-sectional view illustrating ink flows in the ejection unit;
- (25) FIG. 14B is a cross-sectional view illustrating ink flows in the ejection unit;
- (26) FIG. 14C is a cross-sectional view illustrating ink flows in the ejection unit;
- (27) FIG. 15A is a view illustrating a channel configuration of the liquid ejection head;
- (28) FIG. 15B is a view illustrating the channel configuration of the liquid ejection head;
- (29) FIG. 16 is a diagram illustrating a state where main body units of the liquid ejection apparatus and the liquid ejection head are connected;
- (30) FIG. 17A is a cross-sectional view illustrating the vicinity of an ejection port;
- (31) FIG. 17B is a cross-sectional view illustrating the vicinity of the ejection port;
- (32) FIG. 18A is a cross-sectional view illustrating a comparative example of the vicinity of an ejection port;
- (33) FIG. 18B is a cross-sectional view illustrating the comparative example of the vicinity of the ejection port;
- (34) FIG. 19 is a view illustrating a comparative example of an ejection element substrate; and
- (35) FIG. 20 is a diagram illustrating ejection element substrates in ejection modules in a modification.

DESCRIPTION OF THE EMBODIMENTS

(36) A preferred embodiment of the present disclosure will be specifically described with reference to the accompanying drawings. Note that the following embodiment does not limit the contents of the present disclosure, and not all of the combinations of the features described in these embodiments are necessarily essential for the solving means of the present disclosure. Note that identical constituent elements are denoted by the same reference numeral. The present embodiment will be described using an example in which a thermal type ejection element that ejects a liquid by generating a bubble with an electrothermal conversion element is employed as each ejection element that ejects a liquid, but is not limited to this example. The present embodiment is applicable also to liquid ejection heads employing an ejection method in which a liquid is ejected using a piezoelectric element as well as liquid ejection heads employing other ejection methods. Moreover, the pumps, pressure adjustment units, and so on to be described below are not limited to the configurations described in the embodiment and illustrated in the drawings. In the following description, a basic configuration of the present disclosure will be discussed first, and then characteristic features of the present disclosure will be described.

(37) <Liquid Ejection Apparatus>

(38) The present invention is characterized by the extending direction of common supply channels and common collection channels to be described later. To describe this point, the entirety of the liquid ejection apparatus will be described first. FIG. 1A is a view for describing a liquid ejection apparatus, and is an enlarged view of a liquid ejection head of the liquid ejection apparatus and its vicinity. First, a schematic configuration of a liquid ejection apparatus 50 in the present embodiment will be described with reference to FIGS. 1A and 1B. FIG. 1A is a perspective view schematically illustrating the liquid ejection apparatus using the liquid ejection head 1. The liquid ejection apparatus 50 in the present embodiment is configured as a serial inkjet printing apparatus that performs printing on a print medium P by ejecting inks as liquids while scanning the liquid ejection head 1.

(39) The liquid ejection head 1 is mounted on a carriage 60. The carriage 60 reciprocally moves in a main scanning direction (X direction) along a guide shaft 51. The print medium P is conveyed in a sub scanning direction (Y direction) crossing (in this example, perpendicularly crossing) the main

scanning direction by conveyance rollers **55**, **56**, **57**, and **58**. Note that, in drawings to be referred to below, the Z direction represents a vertical direction and crosses (in this example, perpendicularly crosses) a X-Y plane defined by the X direction and the Y direction. The Z direction is also the liquid ejection direction. The liquid ejection head **1** is configured to be attachable to and detachable from the carriage **60** by a user.

(40) The liquid ejection head **1** includes circulation units **54** and a later-described ejection unit **3** (see FIG. 2). While a specific configuration will be described later, the ejection unit **3** includes a plurality of ejection ports and energy generation elements (hereinafter referred to as “ejection elements”) that generate ejection energy for ejecting liquids from the respective ejection ports.

(41) The liquid ejection apparatus **50** also includes ink tanks **2** serving as ink supply sources and external pumps **21**. The inks stored in the ink tanks **2** are supplied to the circulation units **54** through ink supply tubes **59** by driving forces of the external pumps **21**.

(42) The liquid ejection apparatus **50** forms a predetermined image on the print medium P by repeating a printing scan involving performing printing by causing the liquid ejection head **1** mounted on the carriage **60** to eject the inks while moving in the main scanning direction, and a conveyance operation involving conveying the print medium P in the sub scanning direction. Note that the liquid ejection head **1** in the present embodiment is capable of ejecting four types of inks, namely black (B), cyan (C), magenta (M), and yellow (Y) inks, and printing full-color images with these inks. Here, the inks ejectable from the liquid ejection head **1** are not limited to the above four types of inks. The present disclosure is also applicable to liquid ejection heads for ejecting other types of inks. In short, the types and number of inks to be ejected from the liquid ejection head are not limited.

(43) Also, in the liquid ejection apparatus **50**, a cap member (not illustrated) capable of covering the ejection port surface of the liquid ejection head **1** in which its ejection ports are formed is provided at a position separated from the conveyance path for the print medium P in the X direction. The cap member covers the ejection port surface of the liquid ejection head **1** during a non-print operation, and is used for prevention of drying of the ejection ports, protection of the ejection ports, an ink suction operation from the ejection ports, and so on.

(44) Note that the liquid ejection head **1** illustrated in FIG. 1A represents an example where four circulation units **54** corresponding to the four types of inks are included in the liquid ejection head **1**, but it suffices that the circulation units **54** included correspond to the types of liquids to be ejected. Also, a plurality of circulation units **54** may be included for the same type of liquid. In sum, the liquid ejection head **1** can have a configuration including one or more circulation units. The liquid ejection head **1** may be configured not to circulate all of the four types of inks but only circulate at least one of the inks.

(45) FIG. 1B is a block diagram illustrating a control system of the liquid ejection apparatus **50**. A CPU **103** functions as a control unit that controls the operation of each unit of the liquid ejection apparatus **50** based on a program such as a process procedure stored in a ROM **101**. A RAM **102** is used as a work area or the like for the CPU **103** to execute processes. The CPU **103** receives image data from a host apparatus **400** outside the liquid ejection apparatus **50** and controls a head driver **1A** to control the driving of the ejection elements provided in the ejection unit **3**. The CPU **103** also controls drivers for various actuators provided in the liquid ejection apparatus. For example, the CPU **103** controls a motor driver **105A** for a carriage motor **105** for moving the carriage **60**, a motor driver **104A** for a conveyance motor **104** for conveying the print medium P, and the like. Moreover, the CPU **103** controls a pump driver **500A** for later-described circulation pumps **500**, a pump driver **21A** for the external pumps **21**, and the like. Note that FIG. 1B illustrates a configuration in which the image data is received from the host apparatus **400** and processes are performed, but the liquid ejection apparatus **50** may perform the processes regardless of whether data is given from the host apparatus **400**.

(46) <Basic Configuration of Liquid Ejection Head>

(47) FIG. 2 is an exploded perspective view and a top view of the liquid ejection head **1** in the present embodiment. FIGS. 3A and 3B are cross-sectional views of the liquid ejection head **1** illustrated in FIG. 2 along the IIIA-IIIA line. FIG. 3A is a vertical cross-sectional view of the entire liquid ejection head **1**, and FIG. 3B is an enlarged view of an ejection module illustrated in FIG. 3A. A basic configuration of the liquid ejection head **1** in the present embodiment will be described below with reference mainly to FIGS. 2 to 3B and to FIG. 1A as appropriate.

(48) As illustrated in FIG. 2, the liquid ejection head **1** includes the circulation units **54** and the ejection unit **3** for ejecting the inks supplied from the circulation units **54** onto the print medium P. The liquid ejection head **1** in the present embodiment is fixedly supported on the carriage **60** of the liquid ejection apparatus **50** by a positioning unit and electric contacts (not illustrated) which are provided to the carriage **60**. The liquid ejection head **1** performs printing on the print medium P by ejecting the inks while moving along with the carriage **60** in the main scanning direction (X direction) illustrated in FIG. 1A.

(49) The external pumps **21** connected to the ink tanks **2** serving as ink supply sources include the ink supply tubes **59** (see FIG. 1A). A liquid connector (not illustrated) is provided at the tip of each of these ink supply tubes **59**. In the state where the liquid ejection head **1** is mounted to the liquid ejection apparatus **50**, the liquid connectors which are provided at the tips of the ink supply tubes **59** and are inlets through which the liquids are introduced are hermetically connected to liquid connector insertion slots **53a** that are provided on a head housing **53** of the liquid ejection head **1**. As a result, ink supply paths extending from the ink tanks **2** to the liquid ejection head **1** through the external pumps **21** are formed. In the present embodiment, four types of inks are used. Hence, four sets each including an ink tank **2**, an external pump **21**, an ink supply tube **59**, and a circulation unit **54** are provided for the respective inks, and four ink supply paths corresponding to the respective inks are formed independently of each other. As described above, the liquid ejection apparatus **50** in the present embodiment includes ink supply systems to which the inks are supplied from the ink tanks **2** provided outside the liquid ejection head **1**. Note that the liquid ejection apparatus **50** in the present embodiment does not include ink collection systems that collect the inks in the liquid ejection head **1** into the ink tanks **2**. Accordingly, the liquid ejection head **1** includes the liquid connector insertion slots **53a** to connect the ink supply tubes **59** of the ink tanks **2** but does not include connector insertion slots to connect tubes for collecting the inks in the liquid ejection head **1** into the ink tanks **2**. Note that a liquid connector insertion slot **53a** is provided for each ink.

(50) In FIG. 3A, reference signs **54B**, **54C**, **54M**, and **54Y** denote the circulation units for the black, cyan, magenta, and yellow inks, respectively. The circulation units have substantially the same configuration, and each circulation unit will be denoted as “circulation unit **54**” in the present embodiment unless otherwise distinguished.

(51) In FIGS. 2 and 3A, the ejection unit **3** includes two ejection modules **300**, the first support member **4**, the second support member **7**, an electric wiring member (electric wiring tape) **5**, and an electric contact substrate **6**. As illustrated in FIG. 3B, each ejection module **300** includes a silicon substrate **310** with a thickness of 0.5 mm to 1 mm and a plurality of ejection elements **15** provided in one surface of the silicon substrate **310**. The ejection elements **15** in the present embodiment each includes an electrothermal conversion element (heater) that generates thermal energy as ejection energy for ejecting the liquid. Electric power through an electric wiring formed on the silicon substrate **310** by a film forming technique is supplied to each of the ejection elements **15**.

(52) Also, an ejection port forming member **320** is formed on a surface of the silicon substrate **310** (the lower surface in FIG. 3B). In the ejection port forming member **320**, a plurality of pressure chambers **12** corresponding to the plurality of ejection elements **15** and a plurality of ejection ports **13** to eject the inks are formed by a photolithographic technique. Moreover, common supply channels **18** and common collection channels **19** are formed in the silicon substrate **310**.

Furthermore, in the silicon substrate **310**, there are formed supply connection channels **323** through

which the common supply channels **18** and the pressure chambers **12** communicate with one another, and collection connection channels **324** through which the common collection channels **19** and the pressure chambers **12** communicate with one another. In the present embodiment, one ejection module **300** is configured to eject two types of inks. Specifically, in the two ejection modules illustrated in FIG. 3A, the ejection module **300** located on the left side in FIG. 3A ejects the black and cyan inks, and the ejection module **300** located on the right side in FIG. 3A ejects the magenta and yellow inks. Note that this combination is a mere example, and any combination of inks may be employed. The configuration may be such that one ejection module ejects one type of ink or ejects three or more types of inks. The two ejection modules **300** do not have to eject the same number of types of inks. The configuration may be such that only one ejection module **300** is included, or three or more ejection modules **300** are included. Moreover, in the example illustrated in FIGS. 3A and 3B, two ejection port arrays extending in the Y direction are formed for an ink of one color. A pressure chamber **12**, a common supply channel **18**, and a common collection channel **19** are formed for each of the plurality of ejection ports **13** forming each ejection port array. Note that the present disclosure is characterized in the flow direction of the liquid flowing through this pressure chamber **12**. This point will be described later in detail.

(53) Later-described ink supply ports and ink collection ports are formed on the back surface (the upper surface in FIG. 3B) side of the silicon substrate **310**. Through the ink supply ports, the inks are supplied into the plurality of common supply channels **18** from ink supply channels **48**. Through the ink collection ports, the inks are collected into ink collection channels **49** from the plurality of common collection channels **19**.

(54) Note that the ink supply ports and the ink collection ports correspond to openings for supplying and collecting the inks during later-described forward ink circulation, respectively. Specifically, during the forward ink circulation, the inks are supplied from the ink supply ports into the common supply channels **18**, and the inks are collected from the common collection channels **19** into the ink collection ports. Note that ink circulation in which the inks are caused to flow in the opposite direction may also be performed. In this case, the inks are supplied from the above-described ink collection ports into the common collection channels **19**, and the inks are collected from the common supply channels **18** into the ink supply ports.

(55) As illustrated in FIG. 3A, the back surfaces (the upper surfaces in FIG. 3A) of the ejection modules **300** are adhesively fixed to one surface of the first support member **4** (the lower surface in FIG. 3A). The ink supply channels **48** and the ink collection channels **49**, which penetrate from one surface of the first support member **4** to the opposite surface of the first support member **4**, are formed in the first support member **4**. The openings of the ink supply channels **48** on one side communicate with the above-mentioned ink supply ports in the silicon substrate **310**. The openings of the ink collection channels **49** on the one side communicate with the above-mentioned ink collection ports in the silicon substrate **310**. Note that the ink supply channels **48** and the ink collection channels **49** are provided independently for each type of ink.

(56) Also, the second support member **7** having openings **7a** (see FIG. 2) to insert the ejection modules **300** are adhesively fixed to one surface (the lower surface in FIG. 3A) of the first support member **4**. The electric wiring member **5** to be electrically connected to the ejection modules **300** is held on the second support member **7**. The electric wiring member **5** is a member for applying electric signals for ink ejection to the ejection modules **300**. The electric connection parts of the ejection modules **300** and the electric wiring member **5** are sealed with a sealant (not illustrated) to be protected from corrosion by the inks and external impacts.

(57) Also, the electric contact substrate **6** is joined to an end portion **5a** of the electric wiring member **5** (see FIG. 2) by thermocompression bonding with an anisotropic conductive film (not illustrated), and the electric wiring member **5** and the electric contact substrate **6** are electrically connected to each other. The electric contact substrate **6** has external signal input terminals (not illustrated) for receiving electric signals from the liquid ejection apparatus **50**.

(58) Moreover, a joint member **8** (FIG. 3A) is provided between the first support member **4** and the circulation units **54**. In the joint member **8**, a supply port **88** and a collection port **89** are formed for each type of ink. Through the supply ports **88** and the collection ports **89**, the ink supply channels **48** and the ink collection channels **49** in the first support member **4** and channels formed in the circulation units **54** communicate with each other. Incidentally, in FIG. 3A, a supply port **88B** and a collection port **89B** are for the black ink, and a supply port **88C** and a collection port **89C** are for the cyan ink. Moreover, a supply port **88M** and a collection port **89M** are for the magenta ink, and a supply port **88Y** and a collection port **89Y** are for the yellow ink.

(59) Note that the openings at one end of the ink supply channels **48** and the ink collection channels **49** in the first support member **4** have small opening areas matching the ink supply ports and the ink collection ports in the silicon substrate **310**. On the other hand, the openings at the other end of the ink supply channels **48** and the ink collection channels **49** in the first support member **4** have a large shape whose opening area is the same opening area formed in the joint member **8** to match the channels in the circulation units **54**. Employing such a configuration can suppress an increase in channel resistance on the ink collected from each collection channel. Note that the shapes of the openings at one end and the other end of the ink supply channels **48** and the ink collection channels **49** are not limited to the above example.

(60) In the liquid ejection head **1** having the above configuration, the inks supplied to the circulation units **54** pass through the supply ports **88** in the joint member **8** and the ink supply channels **48** in the first support member **4** and flow into the common supply channels **18** from the ink supply ports in the ejection modules **300**. Thereafter, the inks flow from the common supply channels **18** into the pressure chambers **12** through the supply connection channels **323**. Part of the inks flowing into the pressure chambers is ejected from the ejection ports **13** as the ejection elements **15** are driven. The remaining inks not ejected pass through the collection connection channels **324** and the common collection channels **19** from the pressure chambers **12**, and flow from the ink collection ports into the ink collection channels **49** in the first support member **4**. Then, the inks flowing into the ink collection channels **49** flow into the circulation units **54** through the collection ports **89** in the joint member **8** and are collected.

(61) <Constituent Elements of Circulation Units>

(62) FIG. 4 is a schematic external view of one circulation unit **54** for one type of ink used in a printing apparatus in the present embodiment. A filter **110**, the first pressure adjustment unit **120**, the second pressure adjustment unit **150**, and a circulation pump **500** are disposed in the circulation unit **54**. As illustrated in FIGS. 5 and 6, these constituent elements are connected by channels to form a circulation path for supplying and collecting the ink to and from the ejection module **300** in the liquid ejection head **1**.

(63) <Circulation Path in Liquid Ejection Head>

(64) FIG. 5 is a vertical cross-sectional view schematically illustrating the circulation path for one type of ink (ink of one color) formed in the liquid ejection head **1**. The relative positions of the components in FIG. 5 (such as the first pressure adjustment unit **120**, the second pressure adjustment unit **150**, and the circulation pump **500**) are simplified for a clearer description of the circulation path. Thus, the relative positions of the components are different from those of the components in FIG. 16 to be mentioned later. Incidentally, FIG. 6 is a block diagram schematically illustrating the circulation path illustrated in FIG. 5. As illustrated in FIGS. 5 and 6, the first pressure adjustment unit **120** includes the first valve chamber **121** and the first pressure control chamber **122**. The second pressure adjustment unit **150** includes the second valve chamber **151** and the second pressure control chamber **152**. The first pressure adjustment unit **120** is configured such that the controlled pressure therein is higher than that in the second pressure adjustment unit **150**. In the present embodiment, these two pressure adjustment units **120** and **150** are used to implement circulation within a certain pressure range inside the circulation path. Also, the configuration is such that the ink flows through the pressure chambers **12** (ejection elements **15**) at a flow rate

corresponding to the pressure difference between the first pressure adjustment unit **120** and the second pressure adjustment unit **150**. A circulation path in the liquid ejection head **1** and a flow of the ink in the circulation path will be described below with reference to FIGS. **5** and **6**. Note that the arrows in FIGS. **5** and **6** indicate the flow direction of the ink.

(65) First, how the constituent elements in the liquid ejection head **1** are connected will be described.

(66) The external pump **21**, which sends the ink stored in the ink tank **2** (FIG. **6**) disposed outside the liquid ejection head **1** to the liquid ejection head **1**, is connected to the circulation unit **54** through the ink supply tube **59** (FIG. **1**). The filter **110** is disposed in the ink channel located on an upstream side of the circulation unit **54**. The ink supply path located downstream of the filter **110** is connected to the first valve chamber **121** of the first pressure adjustment unit **120**. The first valve chamber **121** communicates with the first pressure control chamber **122** through a communication port **191A** openable and closable by a valve **190A** illustrated in FIG. **5**.

(67) The first pressure control chamber **122** is connected to a supply channel **130**, a bypass channel **160**, and a pump outlet channel **180** of the circulation pump **500**. The supply channel **130** is connected to the common supply channels **18** through the above-mentioned ink supply ports provided in the ejection module **300**. Also, the bypass channel **160** is connected to the second valve chamber **151** provided in the second pressure adjustment unit **150**. The second valve chamber **151** communicates with the second pressure control chamber **152** through a communication port **191B** that is opened and closed by a valve **190B** illustrated in FIG. **5**. Note that FIGS. **5** and **6** illustrate an example where one end of the bypass channel **160** is connected to the first pressure control chamber **122** of the first pressure adjustment unit **120**, and the other end of the bypass channel **160** is connected to the second valve chamber **151** of the second pressure adjustment unit **150**.

However, the one end of the bypass channel **160** may be connected to the supply channel **130**, and the other end of the bypass channel may be connected to the second valve chamber **151**.

(68) The second pressure control chamber **152** is connected to a collection channel **140**. The collection channel **140** is connected to the common collection channels **19** through the above-mentioned ink collection ports provided in the ejection module **300**. Moreover, the second pressure control chamber **152** is connected to the circulation pump **500** through a pump inlet channel **170**. Note that reference sign **170a** in FIG. **5** denotes an inlet port of the pump inlet channel **170**.

(69) Next, the flow of the ink in the liquid ejection head **1** having the above configuration will be described. As illustrated in FIG. **6**, the ink stored in the ink tank **2** is pressurized by the external pump **21** provided in the liquid ejection apparatus **50**, becomes an ink flow at a positive pressure, and is supplied to the circulation unit **54** of the liquid ejection head **1**.

(70) The ink supplied to the circulation unit **54** passes through the filter **110** so that foreign substances such as dust and bubbles are removed. The ink then flows into the first valve chamber **121** provided in the first pressure adjustment unit **120**. The pressure on the ink decreases due to the pressure loss in a case where the ink passes through the filter **110**, but the pressure on the ink is still positive at this point. Thereafter, in a case where the valve **190A** is open, the ink flowing into the first valve chamber **121** passes through the communication port **191A** and flows into the first pressure control chamber **122**. Due to the pressure loss in a case where the ink passes through the communication port **191A**, the pressure on the ink flowing into the first pressure control chamber **122** switches from the positive pressure to a negative pressure.

(71) Next, the flow of the ink in the circulation path will be described. The circulation pump **500** operates such that the ink sucked from the pump inlet channel **170** located upstream of the circulation pump **500** is sent to the pump outlet channel **180** located downstream of the circulation pump **500**. Thus, as the pump is driven, the ink supplied to the first pressure control chamber **122** flows into the supply channel **130** and the bypass channel **160** along with the ink sent from the pump outlet channel **180**. In the present embodiment, while details will be described later, a piezoelectric diaphragm pump using a piezoelectric element attached to a diaphragm as a driving

source is used as a circulation pump capable of sending the liquid. The piezoelectric diaphragm pump is a pump that sends a liquid by inputting a driving voltage to a piezoelectric element to change the volume of a pump chamber and alternatively moving two check valves in response to the changes in pressure.

(72) The ink flowing into the supply channel **130** flows from the ink supply ports in the ejection module **300** into the pressure chambers **12** through the common supply channels **18**. Part of the ink is ejected from the ejection ports **13** as the ejection elements **15** are driven (generate heat). Also, the remaining ink not used in the ejection flows through the pressure chambers **12** and passes through the common collection channels **19**. Thereafter, the ink flows into the collection channel **140** connected to the ejection module **300**. The ink flowing into the collection channel **140** flows into the second pressure control chamber **152** of the second pressure adjustment unit **150**.

(73) On the other hand, the ink flowing from the first pressure control chamber **122** into the bypass channel **160** flows into the second valve chamber **151**, passes through the communication port **191B**, and then flows into the second pressure control chamber **152**. The ink flowing into the second pressure control chamber **152** through the bypass channel **160** and the ink collected from the collection channel **140** are sucked into the circulation pump **500** through the pump inlet channel **170** as the circulation pump **500** is driven. Then, the inks sucked into the circulation pump **500** are sent to the pump outlet channel **180** and flow into the first pressure control chamber **122** again. Thereafter, the ink flowing from the first pressure control chamber **122** into the second pressure control chamber **152** through the supply channel **130** and the ejection module **300** and the ink flowing into the second pressure control chamber **152** through the bypass channel **160** flow into the circulation pump **500**. Then, the inks are sent from the circulation pump **500** to the first pressure control chamber **122**. The ink circulation is performed within the circulation path in this manner.

(74) As described above, in the present embodiment, the liquids can be circulated through the respective circulation paths formed in the liquid ejection head **1** with the circulation pump **500**. This makes it possible to suppress thickening of the inks and deposition of precipitating components of the inks of the color materials in the ejection modules **300**. Accordingly, the excellent fluidity of the inks in the ejection modules **300** and excellent ejection characteristics at the ejection ports can be maintained.

(75) Also, the circulation paths in the present embodiment are configured to be completely within the liquid ejection head **1**. Thus, the length of the circulation paths is significantly short as compared to a case where the inks are circulated between the ink tanks **2** disposed outside the liquid ejection head **1** and the liquid ejection head **1**. Accordingly, the inks can be circulated with small circulation pumps.

(76) Moreover, the configuration is such that only channels for supplying the inks are included as the channels connecting between the liquid ejection head **1** and the ink tanks **2**. In other words, a configuration that does not require channels for collecting the inks from the liquid ejection head **1** into the ink tanks **2** is employed. Accordingly, only ink supply tubes connecting between the ink tanks **2** and the liquid ejection head **1** are needed, and no ink collection tube is required. The inside of the liquid ejection apparatus **50** therefore has a simpler configuration having less tubes. This can downsize the entire apparatus. Moreover, the reduction in the number of tubes reduces the fluctuations in ink pressure due to the swinging of the tubes caused by main scanning of the liquid ejection head **1**. Also, the swinging of the tubes during main scanning of the liquid ejection head **1** increases a driving load on the carriage motor driving the carriage **60**. Hence, the reduction of the number of tubes reduces the driving load of the carriage motor, which makes it possible to simplify the main scanning mechanism including the carriage motor and the like. Furthermore, since the inks do not need to be collected into the ink tanks from the liquid ejection head **1**, the external pumps **21** can be downsized as well. As described above, according to the present embodiment, it is possible to downsize the liquid ejection apparatus **50** and reduce costs.

(77) <Pressure Adjustment Units>

(78) FIGS. 7A to 7C are views illustrating an example of the pressure adjustment units. Configurations and operation of the pressure adjustment units incorporated in the above-described liquid ejection head **1** (first pressure adjustment unit **120** and second pressure adjustment unit **150**) will be described in more detail with reference to FIGS. 7A to 7C. Note that the first pressure adjustment unit **120** and the second pressure adjustment unit **150** have substantially the same configuration. Thus, the following description will be given by taking the first pressure adjustment unit **120** as an example. As for the second pressure adjustment unit **150**, only the reference signs of its portions corresponding to those of the first pressure adjustment unit are presented in FIGS. 7A to 7C. In a case of the second pressure adjustment unit **150**, the first valve chamber **121** and the first pressure control chamber **122** described below should be read as the second valve chamber **151** and the second pressure control chamber **152**, respectively.

(79) The first pressure adjustment unit **120** has the first valve chamber **121** and the first pressure control chamber **122** formed in a cylindrical housing **125**. The first valve chamber **121** and the first pressure control chamber **122** are separated by a partition **123** provided inside the cylindrical housing **125**. However, the first valve chamber **121** communicates with the first pressure control chamber **122** through a communication port **191** formed in the partition **123**. A valve **190**, which switches between allowing communication between the first valve chamber **121** and the first pressure control chamber **122** through the communication port **191** and blocking the communication, is provided in the first valve chamber **121**. The valve **190** is held by a valve spring **200** at a position opposite to the communication port **191**, and has a tight contact configuration to the partition **123** by a biasing force from the valve spring **200**. The valve **190** blocks the ink flow through the communication port **191** by being in tight contact with the partition **123**. Note that the portion of the valve **190** to be in contact with the partition **123** is preferably formed of an elastic member in order to enhance the tightness of the contact with the partition **123**. Also, a valve shaft **190a** to be inserted through the communication port **191** is provided in a protruding manner on a center portion of the valve **190**. By pressing this valve shaft **190a** against the biasing force from the valve spring **200**, the valve **190** gets separated from the partition **123**, thereby allowing the ink to flow through the communication port **191**. In the following, the state where the valve **190** blocks the ink flow through the communication port **191** will be referred to as “closed state”, and the state where the ink can flow through the communication port **191** will be referred to as “open state”.

(80) The opening portion of the cylindrical housing **125** is closed by a flexible member **230** and a pressing plate **210**. These flexible member **230** and pressing plate **210**, the peripheral wall of the housing **125**, and the partition **123** form the first pressure control chamber **122**. The pressing plate **210** is configured to be displaceable with displacement of the flexible member **230**. While the materials of the pressing plate **210** and the flexible member **230** are not particularly limited, for example, the pressing plate **210** can be made as a molded resin component, and the flexible member **230** can be made from a resin film. In this case, the pressing plate **210** can be fixed to the flexible member **230** by thermal welding.

(81) A pressure adjustment spring **220** (biasing member) is provided between the pressing plate **210** and the partition **123**. As illustrated in FIG. 7A, the pressing plate **210** and the flexible member **230** are biased by a biasing force from the pressure adjustment spring **220** in a direction in which the inner volume of the first pressure control chamber **122** increases. Also, as the pressure in the first pressure control chamber **122** decreases, the pressing plate **210** and the flexible member **230** get displaced against the pressure from the pressure adjustment spring **220** in the direction in which the inner volume of the first pressure control chamber **122** decreases. Then, in a case where the inner volume of the first pressure control chamber **122** decreases to a certain volume, the pressing plate **210** abuts the valve shaft **190a** of the valve **190**. As the inner volume of the first pressure control chamber **122** then decreases further, the valve **190** moves with the valve shaft **190a** against the biasing force from the valve spring **200**, thereby being separated from the partition **123**. As a result, the communication port **191** shifts to the open state (the state of FIG. 7B).

(82) In the present embodiment, the connections in the circulation path are set such that the pressure in the first valve chamber **121** in a case where the communication port **191** shifts to the open state is higher than the pressure in the first pressure control chamber **122**. In this way, in a case where the communication port **191** shifts to the open state, the ink flows from the first valve chamber **121** into the first pressure control chamber **122**. The inflow of the ink displaces the flexible member **230** and the pressing plate **210** in the direction in which the inner volume of the first pressure control chamber **122** increases. As a result, the pressing plate **210** gets separated from the valve shaft **190a** of the valve **190**, and the valve **190** is brought into tight contact with the partition **123** by the biasing force from the valve spring **200** so that the communication port **191** shifts to the closed state (the state of FIG. 7C).

(83) As described above, in the first pressure adjustment unit **120** in the present embodiment, in a case where the pressure in the first pressure control chamber **122** decreases to a certain pressure or less (e.g., in a case where the negative pressure becomes strong), the ink flows from the first valve chamber **121** through the communication port **191**. This configuration limits the pressure in the first pressure control chamber **122** from decreasing any further. Accordingly, the pressure in the first pressure control chamber **122** is controlled to be maintained within a certain range.

(84) Next, the pressure in the first pressure control chamber **122** will be described in more detail.

(85) Consider a state where the flexible member **230** and the pressing plate **210** are displaced according to the pressure in the first pressure control chamber **122** as described above so that the pressing plate **210** abuts the valve shaft **190a** and brings the communication port **191** into the open state (the state of FIG. 7B). The relation between the forces acting on the pressing plate **210** at this time is represented by Equation 1 below.

$$P2 \times S2 + F2 + (P1 - P2) \times S1 + F1 = 0 \quad \text{Equation 1}$$

(86) Moreover, Equation 1 is summarized for $P2$ as below.

$$P2 = -(F1 + F2 + P1 \times S1) / (S2 - S1) \quad \text{Equation 2}$$

$P1$: Pressure (gauge pressure) in the first valve chamber **121**
 $P2$: Pressure (gauge pressure) in first pressure control chamber **122**
 $F1$: Spring force of the valve spring **200**
 $F2$: Spring force of the pressure adjustment spring **220**
 $S1$: Pressure reception area of the valve **190**
 $S2$: Pressure reception area of the pressing plate **210**

(87) Here, as for the spring force $F1$ of the valve spring **200** and the spring force $F2$ of the pressure adjustment spring **220**, the direction in which they push the valve **190** and the pressing plate **210** is defined as the forward direction (the leftward direction in FIGS. 7A to 7C). Also, the configuration is such that the pressure $P1$ in the first valve chamber **121** and the pressure $P2$ in the first pressure control chamber **122** satisfy a relation of $P1 \geq P2$.

(88) The pressure $P2$ in the first pressure control chamber **122** when the communication port **191** shifts to the open state is determined by Equation 2 and, since the configuration is such that the relation of $P1 \geq P2$ is satisfied, the ink flows into the first pressure control chamber **122** from the first valve chamber **121** when the communication port **191** shifts to the open state. As a result, the pressure $P2$ in the first pressure control chamber **122** does not decrease any further, and the pressure $P2$ is kept at a pressure within a certain range.

(89) On the other hand, as illustrated in FIG. 7C, the relation between the forces acting on the pressing plate **210** in a case where the pressing plate **210** does not abut on the valve shaft **190a** and the communication port **191** shifts to the closed state is represented by Equation 3 below.

$$P3 \times S3 + F3 = 0 \quad \text{Equation 3}$$

(90) Here, Equation 3 is summarized for $P3$ as below.

$$P3 = F3 / S3 \quad \text{Equation 4}$$

(91) $F3$: Spring force of the pressure adjustment spring **220** in a state where the pressing plate **210** does not abut on the valve shaft **190a**

(92) $P3$: Pressure (gauge pressure) in the first pressure control chamber **122** in the state where the pressing plate **210** does not abut on the valve shaft **190a**

(93) $S3$: Pressure reception area of the pressing plate **210** in a state where the pressing plate **210**

does not abut on the valve shaft **190a**

(94) Here, FIG. 7C illustrates a state where the pressing plate **210** and the flexible member **230** are displaced in the leftward direction in FIG. 7C up to the limit to which they can be displaced. The pressure **P3** in the first pressure control chamber **122**, the spring force **F3** of the pressure adjustment spring **220**, and the pressure reception area **S3** of the pressing plate **210** change depending on the amount of displacement of the pressing plate **210** and the flexible member **230** in displacement to the state of FIG. 7C. Specifically, in a case where the pressing plate **210** and the flexible member **230** are situated on the right side in FIG. 7C relative to themselves in FIG. 7C, the pressure reception area **S3** of the pressing plate **210** is smaller and the spring force **F3** of the pressure adjustment spring **220** is larger. Accordingly, the pressure **P3** in the first pressure control chamber **122** is smaller in accordance with the relation in Equation 4. Thus, with Equations 2 and 4, the pressure in the first pressure control chamber **122** gradually increases (that is, the negative pressure weakens toward a value close to the positive pressure side) in shifting from the state of FIG. 7B to the state of FIG. 7C. Specifically, the pressure in the first pressure control chamber **122** gradually increases while the pressing plate **210** and the flexible member **230** are gradually displaced in the leftward direction from the state where the communication port **191** is in the open state to the state where the inner volume of the first pressure control chamber reaches the limit to which the pressing plate **210** and the flexible member **230** can be displaced. In other words, the negative pressure weakens.

(95) <Circulation Pumps>

(96) Next, a configuration and operation of each circulation pump **500** incorporated in the above liquid ejection head **1** will be described in detail with reference to FIGS. 8A and 8B and FIG. 9.

(97) FIGS. 8A and 8B are external perspective views of the circulation pump **500**. FIG. 8A is an external perspective view illustrating the front side of the circulation pump **500**, and FIG. 8B is an external perspective view illustrating the back side of the circulation pump **500**. An outer shell of the circulation pump **500** includes a pump housing **505** and a cover **507** fixed to the pump housing **505**. The pump housing **505** includes a housing-part main body **505a** and a channel connection member **505b** adhesively fixed to the outer surface of the housing-part main body **505a**. In each of the housing-part main body **505a** and the channel connection member **505b**, a pair of through-holes communicating with each other are formed at two different positions. One of the pair of through-holes provided at one position forms a pump supply hole **501**. The other of the pair of through-holes provided at the other position forms a pump discharge hole **502**. The pump supply hole **501** is connected to the pump inlet channel **170** connected to the second pressure control chamber **152**. The pump discharge hole **502** is connected to the pump outlet channel **180** connected to the first pressure control chamber **122**. The ink supplied from the pump supply hole **501** passes through a later-described pump chamber **503** (see FIG. 9) and is discharged from the pump discharge hole **502**.

(98) FIG. 9 is a cross-sectional view of the circulation pump **500** illustrated in FIG. 8A along the IX-IX line. A diaphragm **506** is joined to the inner surface of the pump housing **505**, and the pump chamber **503** is formed between this diaphragm **506** and a recess formed in the inner surface of the pump housing **505**. The pump chamber **503** communicates with the pump supply hole **501** and the pump discharge hole **502**, which are formed in the pump housing **505**. Also, a check valve **504a** is provided at an intermediate portion of the pump supply hole **501**. A check valve **504b** is provided at an intermediate portion of the pump discharge hole **502**. Specifically, the check valve **504a** is disposed such that a part thereof is movable in the leftward direction in FIG. 9 within a space **512a** formed at an intermediate portion of the pump supply hole **501**. The check valve **504b** is disposed such that a part thereof is movable in the rightward direction in FIG. 9 within a space **512b** formed at an intermediate portion of the pump discharge hole **502**.

(99) As the diaphragm **506** is displaced so as to increase the volume of the pump chamber **503**, the pump chamber **503** is depressurized. In response to this displacement, the check valve **504a** is

separated from the opening of the pump supply hole **501** in the space **512a** (that is, moves in the leftward direction in FIG. **9**). By being separated from the opening of the pump supply hole **501** in the space **512a**, the check valve **504a** shifts to an open state in which the ink is allowed to flow through the pump supply hole **501**. As the diaphragm **506** is displaced so as to reduce the volume of the pump chamber **503**, the pump chamber **503** is pressurized. In response to this displacement, the check valve **504a** comes into tight contact with the wall surface around the opening of the pump supply hole **501**. The check valve **504a** is thus in a closed state in which the check valve **504a** blocks the ink flow through the pump supply hole **501**.

(100) The check valve **504b**, on the other hand, comes into tight contact with the wall surface around an opening in the pump housing **505** as the pump chamber **503** is depressurized, thereby shifting to a closed state in which the check valve **504b** blocks the ink flow through the pump discharge hole **502**. Also, as the pump chamber **503** is pressurized, the check valve **504b** is separated from the opening in the pump housing **505** and moves toward the space **512b** (that is, moves in the rightward direction in FIG. **9**), thereby allowing the ink to flow through the pump discharge hole **502**.

(101) Note that the material of each of the check valves **504a** and **504b** only needs to be one that is deformable according to the pressure in the pump chamber **503**. For example, the material of each of the check valves **504a** and **504b** can be made from an elastic material such as Ethylene-Propylene-Diene Methylene linkage (EPDM) or an elastomer, or a film or thin plate of polypropylene or the like. However, the material is not limited to these.

(102) As described above, the pump chamber **503** is formed by joining the pump housing **505** and the diaphragm **506**. Thus, the pressure in the pump chamber **503** changes as the diaphragm **506** is deformed. For example, in a case where the diaphragm **506** is displaced toward the pump housing **505** (displaced toward the right side in FIG. **9**), thereby reducing the volume of the pump chamber **503**, the pressure in the pump chamber **503** increases. As a result, the check valve **504b** disposed so as to face the pump discharge hole **502** shifts to the open state so that the ink in the pump chamber **503** is discharged. At this time, the check valve **504a** disposed so as to face the pump supply hole **501** is in tight contact with the wall surface around the pump supply hole **501**, thereby suppressing backflow of the ink from the pump chamber **503** into the pump supply hole **501**.

(103) Conversely, in a case where the diaphragm **506** is displaced in the direction in which the pump chamber **503** widens, the pressure in the pump chamber **503** decreases. As a result, the check valve **504a** disposed so as to face the pump supply hole **501** shifts to the open state so that the ink is supplied into the pump chamber **503**. At this time, the check valve **504b** disposed in the pump discharge hole **502** comes into tight contact with the wall surface around an opening formed in the pump housing **505** to close this opening. This suppresses backflow of the ink from the pump discharge hole **502** into the pump chamber **503**.

(104) As described above, in the circulation pump **500**, the ink is sucked and discharged as the diaphragm **506** is deformed and thereby changes the pressure in the pump chamber **503**. At this time, in a case where bubbles have entered the pump chamber **503**, the displacement of the diaphragm **506** changes the pressure in the pump chamber **503** to a lesser extent due to the expansion or shrinkage of the bubbles. Accordingly, the amount of the liquid to be sent decreases. To resolve this phenomenon, the pump chamber **503** is disposed in parallel with gravity so that the bubbles having entered the pump chamber **503** can easily gather in an upper portion of the pump chamber **503**. In addition, the pump discharge hole **502** is disposed higher than the center of the pump chamber **503**. This improves the ease of discharge of bubbles in the pump and thus stabilizes the flow rate.

(105) <Flow of Ink inside Liquid Ejection Head>

(106) FIGS. **10A** to **10E** are diagrams describing a flow of an ink inside the liquid ejection head. The circulation of the ink performed inside the liquid ejection head **1** will be described with reference to FIGS. **10A** to **10E**. The relative positions of the components in FIGS. **10A** to **10E** such

as the first pressure adjustment unit **120**, the second pressure adjustment unit **150**, and the circulation pump **500** are simplified for a clearer description of the ink circulation path. Thus, the relative positions of the components are different from those of the components in FIG. **19** to be mentioned later. FIG. **10A** schematically illustrates the flow of the ink in a case of performing a print operation of performing printing by ejecting the ink from the ejection ports **13**. Note that the arrows in FIG. **10A** indicate the flow of the ink. In the present embodiment, to perform a print operation, both the external pump **21** and the circulation pump **500** start being driven. Incidentally, the external pump **21** and the circulation pump **500** may be driven regardless of whether a print operation is to be performed or not. The external pump **21** and the circulation pump **500** do not have to be driven in conjunction with each other, and may be driven independently of each other. (107) During the print operation, the circulation pump **500** is in an ON state (driven state) so that the ink flowing out of the first pressure control chamber **122** flows into the supply channel **130** and the bypass channel **160**. The ink having flowed into the supply channel **130** passes through the ejection module **300** and then flows into the collection channel **140**. Thereafter, the ink is supplied into the second pressure control chamber **152**.

(108) On the other hand, the ink that flowed into the bypass channel **160** from the first pressure control chamber **122** flows into the second pressure control chamber **152** through the second valve chamber **151**. The ink that flowed into the second pressure control chamber **152** passes through the pump inlet channel **170**, the circulation pump **500**, and the pump outlet channel **180** and then flows into the first pressure control chamber **122** again. At this time, based on the relation in Equation 2 mentioned above, the controlled pressure in the first valve chamber **121** is set higher than the controlled pressure in the first pressure control chamber **122**. Thus, the ink in the first pressure control chamber **122** does not flow into the first valve chamber **121** but is supplied to the ejection module **300** again through the supply channel **130**. The ink that flowed into the ejection module **300** flows into the first pressure control chamber **122** again through the collection channel **140**, the second pressure control chamber **152**, the pump inlet channel **170**, the circulation pump **500**, and the pump outlet channel **180**. Ink circulation that is completely within the liquid ejection head **1** is performed as described above.

(109) In the above ink circulation, the differential pressure between the controlled pressure in the first pressure control chamber **122** and the controlled pressure in the second pressure control chamber **152** determines the amount of circulation (flow rate) of the ink within the ejection module **300**. Moreover, this differential pressure is set to obtain an amount of circulation that can suppress thickening of the ink near the ejection ports in the ejection module **300**. Incidentally, the amount of the ink consumed by the printing is supplied from the ink tank **2** to the first pressure control chamber **122** through the filter **110** and the first valve chamber **121**. How the consumed ink is supplied will now be described in detail. The ink in the circulation path decreases by the amount of the ink consumed by the printing. Accordingly, the pressure in the first pressure control chamber **122** decreases, resulting in decreasing the ink in the first pressure control chamber. As the ink in the first pressure control chamber **122** decreases, the inner volume of the first pressure control chamber **122** decreases accordingly. As this inner volume of the first pressure control chamber **122** decreases, the communication port **191A** shifts to the open state so that the ink is supplied from the first valve chamber **121** to the first pressure control chamber **122**. A pressure loss occurs in this supplied ink as this ink supplied from the first valve chamber **121** passes through the communication port **191A**. As the ink flows into the first pressure control chamber **122**, the positive pressure on the ink switches to a negative pressure. As the ink flows from the first valve chamber **121** into the first pressure control chamber **122**, the pressure in the first pressure control chamber increases. The communication port **191A** shifts to the closed state when the inner volume of the first pressure control chamber increases. As described above, the communication port **191A** repetitively switches between the open state and the closed state according to the ink consumption. Incidentally, the communication port **191A** is kept in the closed state in a case where the ink is not

consumed.

(110) FIG. 10B schematically illustrates the flow of the ink immediately after the print operation is finished and the circulation pump 500 shifts to an OFF state (stop state). At the point when the print operation is finished and the circulation pump 500 shifts to the OFF state, the pressure in the first pressure control chamber 122 and the pressure in the second pressure control chamber 152 are both the controlled pressures used in the print operation. For this reason, the ink moves as illustrated in FIG. 10B according to the differential pressure between the pressure in the first pressure control chamber 122 and the pressure in the second pressure control chamber 152. Specifically, the ink flow from the first pressure control chamber 122 to the ejection module 300 through the supply channel 130 and then to the second pressure control chamber 152 through the collection channel 140 continues to be generated. Moreover, the ink flow from the first pressure control chamber 122 to the second pressure control chamber 152 through the bypass channel 160 and the second valve chamber 151 continues to be generated.

(111) The amount of the ink moved from the first pressure control chamber 122 to the second pressure control chamber 152 by these ink flows is supplied from the ink tank 2 to the first pressure control chamber 122 through the filter 110 and the first valve chamber 121. Accordingly, the inner volume of the first pressure control chamber 122 is maintained constant. According to the relation in Equation 2 mentioned above, the spring force F_1 of the valve spring 200, the spring force F_2 of the pressure adjustment spring 220, the pressure reception area S_1 of the valve 190, and the pressure reception area S_2 of the pressing plate 210 are maintained constant in a case where the inner volume of the first pressure control chamber 122 is constant. Thus, the pressure in the first pressure control chamber 122 is determined depending on the change of the pressure (gauge pressure) P_1 in the first valve chamber 121. In this way, in a case where the pressure P_1 in the first valve chamber 121 does not change, the pressure P_2 in the first pressure control chamber 122 is maintained at the same pressure as the controlled pressure in the print operation.

(112) On the other hand, the pressure in the second pressure control chamber 152 changes with time according to the change in inner volume by the inflow of the ink from the first pressure control chamber 122. Specifically, the pressure in the second pressure control chamber 152 changes according to Equation 2 until the communication port 191 shifts from the state of FIG. 10B to the closed state to allow no communication between the second valve chamber 151 and the second pressure control chamber 152 as illustrated in FIG. 10C. Thereafter, the pressing plate 210 does not abut on the valve shaft 190a so that the communication port 191 shifts to the closed state. Then, as illustrated in FIG. 10D, the ink flows from the collection channel 140 into the second pressure control chamber 152. This inflow of the ink displaces the pressing plate 210 and the flexible member 230. The pressure in the second pressure control chamber 152 changes according to Equation 4. Specifically, the pressure increases until the inner volume of the second pressure control chamber 152 reaches the maximum.

(113) Note that, once the state of FIG. 10C is reached, there is no more ink flow from the first pressure control chamber 122 into the second pressure control chamber 152 through the bypass channel 160 and the second valve chamber 151. Thus, the ink flow to the second pressure control chamber 152 through the collection channel 140 is only generated after the ink in the first pressure control chamber 122 is supplied to the ejection module 300 through the supply channel 130. As mentioned above, the ink moves from the first pressure control chamber 122 to the second pressure control chamber 152 according to the differential pressure between the pressure in the first pressure control chamber 122 and the pressure in the second pressure control chamber 152. Thus, in a case where the pressure in the second pressure control chamber 152 becomes equal to the pressure in the first pressure control chamber 122, the ink stops moving.

(114) Also, in the state where the pressure in the second pressure control chamber 152 is equal to the pressure in the first pressure control chamber 122, the second pressure control chamber 152 expands to the state illustrated in FIG. 10D. In a case where the second pressure control chamber

152 expands as illustrated in FIG. **10D**, a reservoir portion capable of holding the ink is formed in the second pressure control chamber **152**. Note that the transition to the state of FIG. **10D** after stopping the circulation pump **500** takes about 1 minute to 2 minutes. The time may vary depending on the shapes and sizes of the channels and properties of the ink. As the circulation pump **500** is driven in the state where the ink is held in the reservoir portion as illustrated in FIG. **10D**, the ink in the reservoir portion is supplied to the first pressure control chamber **122** by the circulation pump **500**. Accordingly, as illustrated in FIG. **10E**, the amount of the ink in the first pressure control chamber **122** increases so that the flexible member **230** and the pressing plate **210** are displaced in the expanding direction. Then, as the circulation pump **500** continues to be driven, the state inside the circulation path changes to the state illustrated in FIG. **10A**.

(115) Note that, in the above description, FIG. **10A** has been described as an example of the ink circulation during a print operation. However, the ink may be circulated without a print operation, as mentioned above. Even in this case, the ink flows as illustrated in FIGS. **10A** to **10E** in response to the driving and stopping of the circulation pump **500**.

(116) Also, as described above, in the present embodiment, an example in which the communication port **191B** in the second pressure adjustment unit **150** shifts to the open state in a case where the ink is circulated by driving the circulation pump **500**, and shifts to the closed state in a case where the ink circulation stops, has been used. However, the present embodiment is not limited to this example. The controlled pressure may be set such that the communication port **191B** in the second pressure adjustment unit **150** is in the closed state even in a case where the ink is circulated by driving the circulation pump **500**. This will be specifically described below along with the function of the bypass channel **160**.

(117) The bypass channel **160** connecting between the first pressure adjustment unit **120** and the second pressure adjustment unit **150** is provided in order that the ejection module **300** can avoid the effect of the strong negative pressure, for example, in a case where the negative pressure generated inside the circulation path becomes stronger than a preset value. The bypass channel **160** is also provided in order to supply the ink to the pressure chambers **12** from both the supply channel **130** and the collection channel **140**.

(118) First, a description will be given of an example of avoiding the effect of the negative pressure becoming stronger than the preset value on the ejection module **300** by providing the bypass channel **160**. For example, a change in environmental temperature sometimes changes a property (e.g., viscosity) of the ink. As the viscosity of the ink changes, the pressure loss within the circulation path changes as well. For example, as the viscosity of the ink decreases, the amount of pressure loss within the circulation path decreases. As a result, the flow rate of the circulation pump **500** driven at a constant driving amount increases, and the flow rate through the ejection module **300** increases. Here, the ejection module **300** is kept at a constant temperature by a temperature adjustment mechanism (not illustrated). Hence, the viscosity of the ink inside the ejection module **300** is maintained constant even if the environmental temperature changes. The viscosity of the ink inside the ejection module **300** remains unchanged whereas the flow rate of the ink flowing through the ejection module **300** increases, and therefore the negative pressure in the ejection module **300** becomes accordingly stronger due to flow resistance. If the negative pressure in the ejection module **300** becomes stronger than the preset value as described above, there is a possibility that the menisci in the ejection ports **13** may break and the ambient air may be taken into the circulation path, which may lead to a failure to perform normal ejection. Also, even if the menisci do not break, there is still a possibility that the negative pressure in the pressure chambers **12** may become stronger than a predetermined level and affect the ejection.

(119) For these reasons, in the present embodiment, the bypass channel **160** is formed in the circulation path. By providing the bypass channel **160**, the ink flows through the bypass channel **160** in a case where the negative pressure is stronger than the preset value. Thus, the pressure in the ejection module **300** is kept constant. Thus, for example, the controlled pressure may be set such

that the communication port **191B** in the second pressure adjustment unit **150** is maintained in the closed state even in a case where the circulation pump **500** is driven. Moreover, the controlled pressure in the second pressure adjustment unit **150** may be set such that the communication port **191B** in the second pressure adjustment unit **150** shifts to the open state in a case where the negative pressure becomes stronger than the preset value. In other words, the communication port **191B** may be in the closed state in a case where the circulation pump **500** is driven as long as the menisci do not collapse or a predetermined negative pressure is maintained even if the flow rate of the pump changes due to the change in viscosity caused by an environmental change or the like.

(120) Next, a description will be given of an example where the bypass channel **160** is provided in order to supply the ink to the pressure chambers **12** from both the supply channel **130** and the collection channel **140**. The pressure in the circulation path may fluctuate due to the ejection operations of the ejection elements **15**. This is because the ejection operations generate a force that draws the ink into the pressure chambers.

(121) In the following, a description will be given of the fact that the ink to be supplied to the pressure chambers **12** is supplied from both the supply channel **130** side and the collection channel **140** side in a case of continuing high-duty printing. While the definition of “duty” may vary depending on various conditions, in the following, a state where a 1200 dpi grid cell is printed with a single 4 pl ink droplet will be considered 100%. “High-duty printing” is, for example, printing performed at a duty of 100%.

(122) In a case of continuing high-duty printing, the amount of the ink flowing from the pressure chambers **12** into the second pressure control chamber **152** through the collection channel **140** decreases. On the other hand, the circulation pump **500** causes the ink to flow out in a constant amount. This breaks the balance between the inflow into and the outflow from the second pressure control chamber **152**. Consequently, the ink inside the second pressure control chamber **152** decreases and the negative pressure in the second pressure control chamber **152** becomes stronger so that the second pressure control chamber **152** shrinks. As the negative pressure in the second pressure control chamber **152** becomes stronger, the amount of inflow of the ink into the second pressure control chamber **152** through the bypass channel **160** increases, and the second pressure control chamber **152** becomes stable in the state where the outflow and the inflow are balanced. Thus, the negative pressure in the second pressure control chamber **152** becomes stronger according to the duty. Also, as mentioned above, under the configuration in which the communication port **191B** is in the closed state in a case where the circulation pump **500** is driven, the communication port **191B** shifts to the open state depending on the duty so that the ink flows from the bypass channel **160** into the second pressure control chamber **152**.

(123) Moreover, as high-duty printing is continued further, the amount of inflow into the second pressure control chamber **152** from the pressure chambers **12** through the collection channel **140** decreases and conversely the amount of inflow into the second pressure control chamber **152** from the communication port **191B** through the bypass channel **160** increases. As this state progresses further, the amount of the ink flowing into the second pressure control chamber **152** from the pressure chambers **12** through the collection channel **140** reaches zero so that the ink flowing from the communication port **191B** is the entire ink flowing out into the circulation pump **500**. As this state progresses further, the ink backs up from the second pressure control chamber **152** into the pressure chambers **12** through the collection channel **140**. In this state, the ink flowing from the second pressure control chamber **152** into the circulation pump **500** and the ink flowing from the second pressure control chamber **152** into the pressure chambers **12** will flow from the communication port **191B** into the second pressure control chamber **152** through the bypass channel **160**. In this case, the ink from the supply channel **130** and the ink from the collection channel **140** are filled into the pressure chambers **12** and ejected therefrom.

(124) Note that this ink backflow that occurs in a case where the printing duty is high is a phenomenon that occurs due to the installation of the bypass channel **160**. Also, as described

above, an example has been described in which the communication port **191B** in the second pressure adjustment unit shifts to the open state for the backflow of the ink. However, the backflow of the ink may also occur in the state where the communication port **191B** in the second pressure adjustment unit is in the open state. Moreover, in a configuration without the second pressure adjustment unit, the above backflow of the ink can also occur by installing the bypass channel **160**.
(125) <Configuration of Ejection Unit>

(126) FIGS. **11A** and **11B** are schematic views illustrating a circulation path for an ink of one color in the ejection unit **3** in the present embodiment. FIG. **11A** is an exploded perspective view of the ejection unit **3** as seen from the first support member **4** side. FIG. **11B** is an exploded perspective view of the ejection unit **3** as seen from the ejection module **300** side. Note that the arrows denoted as “IN” and “OUT” in FIGS. **11A** and **11B** indicate the ink flow, and the ink flow will be described only for one color, but the inks of the other colors flow similarly. Moreover, in FIGS. **11A** and **11B**, illustration of the second support member **7** and the electric wiring member **5** is omitted, and description of them is also omitted in the following description of the configuration of the ejection unit. Moreover, as for the first support member **4** in FIG. **11A**, a cross section along the line XI-XI in FIG. **3A** is illustrated. Each ejection module **300** includes an ejection element substrate **340** and an opening plate **330**. FIG. **12** is a view illustrating the opening plate **330**. FIG. **13** is a view illustrating the ejection element substrate **340**.

(127) The ejection unit **3** is supplied with an ink from each circulation unit **54** through the joint member **8** (see FIG. **3A**). An ink path for an ink to return to the joint member **8** after passing the joint member **8** will now be described. Note that illustration of the joint member **8** is omitted in drawings to be mentioned below.

(128) Each ejection module **300** includes the ejection element substrate **340** and the opening plate **330**, which are the silicon substrate **310**, and further includes the ejection port forming member **320**. The ejection element substrate **340**, the opening plate **330**, and the ejection port forming member **320** form the ejection module **300** by being stacked and joined such that each ink's channels communicate with each other. The ejection module **300** is supported on the first support member **4**. The ejection unit **3** is formed by supporting each ejection module **300** on the first support member **4**. The ejection element substrate **340** includes the ejection port forming member **320**, and the ejection port forming member **320** includes a plurality of ejection port arrays each being a plurality of ejection ports **13** forming a line. Part of the ink supplied through ink channels in the ejection module **300** is ejected from the ejection ports **13**. The ink not ejected is collected through ink channels in the ejection module **300**.

(129) As illustrated in FIGS. **11A** and **11B** and FIG. **12**, the opening plate **330** includes a plurality of arrayed ink supply ports **311** and a plurality of arrayed ink collection ports **312**. As illustrated in FIG. **13** and FIGS. **14A** to **14C**, the ejection element substrate **340** includes a plurality of arrayed supply connection channels **323** and a plurality of arrayed collection connection channels **324**. The ejection element substrate **340** further includes the common supply channels **18** communicating with the plurality of supply connection channels **323** and the common collection channels **19** communicating with the plurality of collection connection channels **324**. The ink supply channels **48** and the ink collection channels **49** (see FIGS. **3A** and **3B**) disposed in the first support member **4** and the channels disposed in each ejection module **300** communicate with each other to form the ink channels inside the ejection unit **3**. Support member supply ports **211** are openings in cross section forming the ink supply channels **48**. Support member collection ports **212** are openings in cross section forming the ink collection channels **49**.

(130) The ink to be supplied to the ejection unit **3** is supplied from the circulation unit **54** (see FIG. **3A**) side to the ink supply channels **48** (see FIG. **3A**) in the first support member **4**.

(131) The ink that flowed through the support member supply ports **211** in the ink supply channels **48** is supplied to the common supply channels **18** in the ejection element substrate **340** through the ink supply channels **48** (see FIG. **3A**) and the ink supply ports **311** in the opening plate **330**, and

enters the supply connection channels **323**. The channels up to this point are the supply-side channels. Thereafter, the ink passes through the pressure chambers **12** (see FIG. **3B**) in the ejection port forming member **320** and flows into the collection connection channels **324** of the collection-side channels. Details of the ink flow in the pressure chambers **12** will be described below.

(132) In the collection-side channels, the ink entered the collection connection channels **324** flows into the common collection channels **19**. Thereafter, the ink flows from the common collection channels **19** into the ink collection channels **49** in the first support member **4** through the ink collection ports **312** in the opening plate **330**, and is collected into the circulation unit **54** through the support member collection ports **212**.

(133) Regions of the opening plate **330** where the ink supply ports **311** or the ink collection ports **312** are not present correspond to regions of the first support member **4** for separating the support member supply ports **211** and the support member collection ports **212**. Also, the first support member **4** does not have openings at these regions. Such regions are used as bonding regions in a case of bonding the ejection module **300** and the first support member **4**.

(134) In FIG. **12**, a plurality of arrays of openings arranged along the X direction are provided side by side in the Y direction in the opening plate **330**, and the openings for supply (IN) and the openings for collection (OUT) are arranged alternately in the Y direction while being shifted from each other by a half pitch in the X direction. In FIG. **13**, in the ejection element substrate **340**, the common supply channels **18** communicating with the plurality of supply connection channels **323** arrayed in the Y direction and the common collection channels **19** communicating with the plurality of collection connection channels **324** arrayed in the Y direction are arrayed alternately in the X direction. The common supply channels **18** and the common collection channels **19** are separated by the ink type. Moreover, the number of ejection port arrays for each color determines the numbers of common supply channels **18** and common collection channels **19** to be disposed. Also, the number of the disposed supply connection channels **323** and the number of the disposed collection connection channels **324** corresponds to the number of ejection ports **13**. Note that a one-to-one correspondence is not necessarily essential, and a single supply connection channel **323** and a single collection connection channel **324** may correspond to a plurality of ejection ports **13**.

(135) Each ejection module **300** is formed by stacking and joining the opening plate **330** and the ejection element substrate **340** as above such that each ink's channels communicate with each other, and is supported on the first support member **4**. As a result, ink channels including the supply channels and the collection channels as above are formed.

(136) FIGS. **14A** to **14C** are cross-sectional views illustrating ink flows at different portions of the ejection unit **3**. FIG. **14A** is a cross section taken along the line XIVA-XIVA in FIG. **11A**, and illustrates a cross section of a portion of the ejection unit **3** where ink supply channels **48** and ink supply ports **311** communicate with each other. FIG. **14B** is a cross section taken along the line XIVB-XIVB in FIG. **11A**, and illustrates a cross section of a portion of the ejection unit **3** where ink collection channels **49** and ink collection ports **312** communicate with each other. Also, FIG. **14C** is a cross section taken along the line XIVC-XIVC in FIG. **11A**, and illustrates a cross section of a portion where the ink supply ports **311** and the ink collection ports **312** do not communicate with channels in the first support member **4**.

(137) As illustrated in FIG. **14A**, the supply channels for supplying the inks supply the inks from the portions where the ink supply channels **48** in the first support member **4** and the ink supply ports **311** in the opening plate **330** overlap and communicate with each other. Moreover, as illustrated in FIG. **14B**, the collection channels for collecting the inks collect the inks from the portions where the ink collection channels **49** in the first support member **4** and the ink collection ports **312** in the opening plate **330** overlap and communicate with each other. Furthermore, as illustrated in FIG. **14C**, the ejection unit **3** locally has regions where no opening is provided in the opening plate **330**. At such regions, the inks are neither supplied or collected between the ejection element substrate **340** and the first support member **4**. The inks are supplied at the regions where

the ink supply ports **311** are provided, as illustrated in FIG. **14A**. The inks are collected at regions where the ink collection ports **312** are provided, as illustrated in FIG. **14B**. Note that the present embodiment has been described by taking the configuration using the opening plate **330** as an example, but a configuration not using the opening plate **330** may be employed. For example, the configuration in which channels corresponding to the ink supply channels **48** and the ink collection channels **49** are formed in the first support member **4**, and the ejection element substrate **340** is joined to the first support member **4** may be employed.

(138) FIGS. **15A** and **15B** are views illustrating a channel configuration of the liquid ejection head **1** for the inks of the three colors of cyan (C), magenta (M), and yellow (Y). In the liquid ejection head **1**, a circulation channel is provided for each ink type as illustrated in FIG. **15A**. The pressure chambers **12** are provided along the X direction, which is the main scanning direction of the liquid ejection head **1**. Also, as illustrated in FIG. **15B**, the common supply channels **18** and the common collection channels **19** are provided along the ejection port arrays, which are arrays of ejection ports **13**. The common supply channels **18** and the common collection channels **19** are provided so as to extend in the Y direction with the ejection port arrays therebetween.

(139) <Connection Between the Main Body Part and the Liquid Ejection Head>

(140) FIG. **16** is schematic configuration diagrams illustrating in more detail the connection state between the ink tank **2** and the external pump **21** provided in a main body part of the liquid ejection apparatus **50** and the liquid ejection head **1**, and placement of circulation pump in the present embodiment. The liquid ejection apparatus **50** in the present embodiment has such a configuration that only the liquid ejection head **1** can be easily replaced in a case where a trouble occurs in the liquid ejection head **1**. Specifically, the liquid ejection apparatus **50** in the present embodiment has the liquid connection parts **700** in which the respective ink supply tubes **59** connected to the respective external pumps **21**, and the liquid ejection head **1** can be easily connected to and disconnected from each other. This enables only the liquid ejection head **1** to be easily attached to and detached from the liquid ejection apparatus **50**.

(141) As illustrated in FIG. **16**, liquid connection part **700** has a liquid connector insertion slot **53a** which is provided in a protruding manner on the head housing **53** of the liquid ejection head **1**, and a cylindrical liquid connector **59a** into which this liquid connector insertion slot **53a** is insertable. The liquid connector insertion slot **53a** is fluidly connected to the ink supply channel formed in the liquid ejection head **1**, and is connected to the first pressure adjustment unit **120** through the filter **110** mentioned above. The liquid connector **59a** is disposed at the tip of the ink supply tube **59** connected to the external pump **21**, which supplies the ink in the ink tank **2** to the liquid ejection head **1** by pressurization.

(142) As described above, the liquid ejection head **1** illustrated in FIG. **16** is available to be easily attached, detached, and replaced of the liquid ejection head **1** by the liquid connection parts **700**. However, there is a possibility that the ink supplied under pressure by an external pump **21** may leak from the liquid connection part **700**, when the sealing performance between the liquid connector insertion slot **53a** and the liquid connector **59a** is deteriorated. If the leaked ink adheres to the circulation pump **500** or the like, there is a possibility that a malfunction will occur in the electrical system. Therefore, in this embodiment, the circulation pumps and the like are arranged as follows.

(143) <Placement of Circulation Pumps, etc.>

(144) As illustrated in FIG. **16**, in the present embodiment, in order to avoid attachment of the ink leaking from the liquid connection part **700** to the circulation pump **500**, the circulation pump **500** is disposed higher than the liquid connection part **700** in the direction of gravity. Specifically, the circulation pump **500** is disposed higher than the liquid connector insertion slot **53a**, which is a liquid inlet in the liquid ejection head **1**, in the direction of gravity. Moreover, the circulation pump **500** is disposed at a position where the constituent members of the liquid connection part **700** are not in contact with the circulation pump **500**. In this way, even if the ink leaks from the liquid

connection part **700**, the ink flows in a horizontal direction which is the opening direction of the opening of the liquid connector **59a** or downward in the direction of gravity. This prevents the ink from reaching the circulation pump **500** located higher in the direction of gravity. Moreover, disposing the circulation pump **500** at a position separated from the liquid connection part **700** also reduces the possibility of the ink reaching the circulation pump **500** through members.

(145) In addition, an electric connection part **515** electrically connecting between the circulation pump **500** and the electric contact substrate **6** through a flexible wiring member **514** is provided higher than the liquid connection part **700** in the direction of gravity. Thus, even in a case where the ink leaks from the liquid connection part **700**, the possibility of that ink causing an electrical trouble is reduced.

(146) Further, in the present embodiment, since a wall portion **53b** of the head housing **53** is provided, even if ink is ejected from the opening **59b** of the liquid connection part **700**, the ink is blocked and the possibility of the ink reaching the circulation pump **500** and the electrical connection part **515** can be reduced.

(147) Characteristic features of the present embodiment will be described below.

(148) FIGS. **17A** and **17B** are cross-sectional views illustrating the vicinity of an ejection port **13** in an ejection module **300**. FIGS. **18A** and **18B** are cross-sectional views illustrating an ejection module with a configuration as a comparative example in which the common supply channels **18** and the common collection channels **19** are widened in the X direction. Note that the bold arrows illustrated in the common supply channel **18** and the common collection channel **19** in FIGS. **17A** and **17B** and FIGS. **18A** and **18B** indicate the oscillating movement of an ink which occurs with the configuration using the serial liquid ejection apparatus **50**.

(149) In the present disclosure, each common supply channel **18** and each common collection channel **19** extend in a direction crossing the main scanning direction (X direction) and the liquid ejection direction (the y direction in FIGS. **17A** and **17B**). Here, the channels extend in the vertical direction (Y direction). With such an arrangement, their channel widths can be made small in the main scanning direction (X direction). This can reduce the effect of the inertial force exerted in the direction opposite to the carriage scanning direction as illustrated in FIGS. **17A** and **17B** (the black bold arrows in FIGS. **17A** and **17B**) on the oscillating movement of the ink. Accordingly, the effect of the oscillating movement on the ejection can be reduced. The extending direction of the common supply channel **18** and the common collection channel **19** is preferably within a range of ± 45 degrees with respect to the main scanning direction (X direction) and the liquid ejection direction (y direction). It is more preferable that the common supply channel **18** and the common collection channel **19** both extend in a direction perpendicularly crossing the main scanning direction (X direction) and the liquid ejection direction (y direction).

(150) Also, as mentioned above, in this configuration, the common supply channel **18** and the common collection channel **19** are separate channels. On the common supply channel **18**, there are supply connection channels **323** through which the ink is supplied to the ejection ports **13**. On the common collection channel **19**, there are collection connection channels **324** through which the ink is collected from the ejection ports **13**. In other words, the ejection ports **13** are present in paths connecting the supply connection channels **323** and the collection connection channels **324**. Thus, at portions of the pressure chambers **12** in the vicinities of the ejection ports **13**, ink flows are generated which flow from the supply connection channel **323** side to the collection connection channel **324** side. The circulation efficiency is significantly good. Also, with such a configuration, even while the carriage is reciprocally moved, ink flows are always generated in the vicinities of the ejection ports **13**, so that the ink circulation is maintained. Thus, the ink inside the pressure chambers **12**, which is most susceptible to ink evaporation from the ejection ports **13**, is always kept fresh. Also, since the two channels, namely the common supply channel **18** and the common collection channel **19**, communicate with the pressure chambers **12**, the ink can be supplied from both channels in a case where it is necessary to perform printing with a high flow rate. That is, in

addition to an advantage in circulation, there is also an advantage that a high flow rate can be handled, as compared to a case where only the supply and collection are implemented through a single channel.

(151) Also, by disposing the common supply channel **18** and the common collection channel **19** at positions overlapping each other in the main scanning direction (X direction), the oscillating movement of the ink during main scanning is substantially the same on the common supply channel **18** side and the common collection channel **19** side as illustrated in FIGS. **17A** and **17B** at any positions in the direction along the ejection port arrays. Thus, the pressure differences generated in the vicinities of the ejection ports **13** between the common supply channel **18** side and the common collection channel **19** side do not greatly vary from each other. Moreover, the common supply channel **18** and the common collection channel **19** may be as close as possible in the X direction. In this way, there is less likely to be a difference in the effect of the oscillating movement of the ink. The gap between the channels is preferably 75 μm or more and 100 μm or less.

(152) Also, regarding the cross-sectional shapes of the common supply channel **18** and the common collection channel **19**, each channel is more preferably elongated in the Z direction. One reason is to increase the cross-sectional area in order to reduce the channel pressure drop. If each channel is widened in the main scanning direction (X direction), the distance between the colors must be widened. This may lower the printing efficiency. Also, if each channel is widened in the main scanning direction (X direction) as illustrated in FIGS. **18A** and **18B**, the ink will receive an inertial force in the main scanning direction to a greater extent. Accordingly, the effect of the oscillating movement during main scanning will be greater. Both the common supply channel **18** and the common collection channel **19** are preferably elongated in the Z direction as illustrated in FIGS. **17A** and **17B**.

(153) FIG. **19** is a view illustrating an ejection element substrate **340** as a comparative example. Note that illustration of the supply connection channels **323** and the collection connection channels **324** is omitted in FIG. **19**. The inks having passed through the pressure chambers flow into the common collection channels **19**. Hence, the temperature of the inks present in the common collection channels **19** is higher than the temperature of the inks in the common supply channels **18**. Here, if the common supply channels **18**, which are lower in temperature than the common collection channels **19**, are situated close to the common collection channels **19**, the temperature of the entire ejection module is kept from rising at least in the vicinities of them. In contrast, if there is a portion where only the common collection channels **19** are present (e.g., a portion a in FIG. **19**), the temperature may locally rise, thereby causing temperature unevenness within the ejection module. This may affect the ejection. For this reason, it is preferable that the common supply channels **18** and the common collection channels **19** have the same length and be present at positions overlapping each other in the main scanning direction (X direction).

(154) As described above, in the present embodiment, the common supply channels **18** and the common collection channels **19** are provided as separate channels, and the pressure chambers **12** are provided in a corresponding relation with the ejection ports **13**. The inks supplied from the common supply channels **18** are supplied into the pressure chambers **12** through the supply connection channels **323**, and are collected from the pressure chambers **12** into the common collection channels **19** through the collection connection channels **324**. Also, the common supply channels **18** and the common collection channels **19** both extend in a direction crossing the main scanning direction (X direction) and the liquid ejection direction (y direction).

(155) With such a configuration, a pressure difference is easily generated in each pressure chamber **12** between the supply side and the collection side. Accordingly, the efficiency of the ink circulation at each pressure chamber **12** is high. Also, a large part of the ink in the pressure chamber **12**, through which the ink is circulated, is ejected from the ejection port **13** as the ejection element **15** provided in the pressure chamber **12** is driven. This enables efficient replacement of the ink in the ejection port **13**.

(156) As described above, in the liquid ejection head **1** in the present embodiment, the common supply channels **18** and the common collection channels **19** are provided as separate channels and both are connected to the pressure chambers **12**. In this way, it is possible to suppress a deterioration in the ink circulation efficiency in the vicinities of the ejection ports. Note that the vicinities of the ejection ports here are regions including the ejection ports **13** and the pressure chambers **12**.

Other Embodiments

(157) FIG. **20** is a view illustrating an arrangement of ejection element substrates **340** in ejection modules **300** in another embodiment. The ejection modules **300** in the present embodiment are disposed differently from the above-described embodiment. The plurality of ejection modules **300** are disposed in the same plane so as to partly overlap each other in the X direction. In other words, their ejection element substrates **340** are disposed so as to be offset from each other in the Y direction as illustrated in FIG. **20**. With the ejection modules **300** disposed so as to be offset from each other in the Y direction as described above, the inks can be ejected as if the ejection port arrays are lengthened. Accordingly, the width of the inks to be ejected in a single scan can be widened. FIG. **20** illustrates the ejection element substrates **340** in two ejection modules **300**. However, the arrangement is not limited to this. Specifically, two or more ejection modules **300** may be disposed so as to be offset from each other in the X direction. In this way, the inks can be ejected even wider.

(158) Note that the common supply channels **18** and the common collection channels **19** in the plurality of ejection modules **300** do not have to be disposed at positions overlapping each other in the X direction. Specifically, it suffices that the common supply channels **18** and the common collection channels **19** in the same ejection module **300** be disposed at positions overlapping each other in the X direction. In this way, it is possible to reduce the effect of the oscillating movement on the ejection ports **13** and the effect of the temperature unevenness.

(159) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(160) This application claims the benefit of Japanese Patent Application No. 2021-205258 filed Dec. 17, 2021, which is hereby incorporated by reference herein in its entirety.

Claims

1. A liquid ejection head comprising: an ejection module configured to eject a liquid in an ejection direction, and including: a plurality of ejection ports, a plurality of energy generation elements configured to eject the liquid respectively from the plurality of ejection ports, a plurality of pressure chambers communicating respectively with the plurality of ejection ports, a plurality of supply connection channels provided in a corresponding relation with the pressure chambers, connected to the pressure chambers, and configured to supply the liquid respectively to the plurality of pressure chambers, a plurality of common supply channels connected to the plurality of supply connection channels, a plurality of collection connection channels provided in a corresponding relation with the pressure chambers, connected to the pressure chambers, and configured to collect the liquid respectively from the plurality of pressure chambers, and a plurality of common collection channels connected to the plurality of collection connection channels; and circulation units each having a circulation pump and a pressure adjustment unit and configured to circulate the liquid by supplying the liquid to the plurality of common supply channels of the ejection module and collecting the liquid from the plurality of common collection channels of the ejection module, wherein the liquid ejection head is capable of moving in a main scanning direction, the common supply channels and the common collection channels extend in a direction crossing the main scanning direction and the

ejection direction, the circulation pump is located inside the liquid ejection head, and the ejection module is connected to two or more circulation units.

2. The liquid ejection head according to claim 1, wherein in the ejection module, the plurality of ejection ports form an ejection port array in an ejection port surface, the ejection port array forming a line in a direction crossing the main scanning direction, and the common supply channels and the common collection channels are provided along the ejection port array with the ejection port array therebetween as viewed in the ejection direction.

3. The liquid ejection head according to claim 2, wherein, in a cross section taken along a plane including the main scanning direction and crossing the ejection port surface, the common supply channels and the common collection channels have a width in the main scanning direction that is shorter than a height thereof in a vertical direction including the ejection direction.

4. The liquid ejection head according to claim 2, wherein a plurality of the ejection port arrays are formed in the ejection port surface, and the common supply channels and the common collection channels are disposed alternately in the main scanning direction.

5. The liquid ejection head according to claim 1, wherein the common supply channels and the common collection channels are provided so as to substantially overlap each other in the main scanning direction.

6. The liquid ejection head according to claim 1, wherein the pressure chambers are provided so as to extend in the main scanning direction.

7. The liquid ejection head according to claim 1, wherein a plurality of the ejection modules are included in a same plane, and the plurality of ejection modules are disposed so as to partly overlap each other in the main scanning direction.

8. The liquid ejection head according to claim 1, wherein the liquid ejection head is configured to be capable of supplying the liquid to the pressure chambers from the collection channels.

9. The liquid ejection head according to claim 1, wherein the liquid is circulated through a circulation path formed in the liquid ejection head with the circulation pump.

10. The liquid ejection head according to claim 1, wherein the pressure adjustment unit comprises a first pressure adjustment unit configured to supply the liquid to the ejection module and a second pressure adjustment unit configured to collect the liquid from the ejection module.

11. A liquid ejection apparatus comprising: a liquid ejection unit configured to eject a liquid in an ejection direction; and a movement unit configured to move the liquid ejection unit in a main scanning direction, wherein the liquid ejection unit includes: an ejection module having a plurality of ejection ports through each of which the liquid is ejectable with operation of an energy generation element, and circulation units each having a circulation pump and a pressure adjustment unit and configured to circulate the liquid by supplying the liquid to the ejection module and collecting the liquid from the ejection module, the ejection module includes: pressure chambers communicating with the ejection ports, a plurality of supply connection channels provided in a corresponding relation with the pressure chambers, connected to the pressure chambers, and configured to supply the liquid respectively to the plurality of pressure chambers, a plurality of common supply channels connected to the plurality of supply connection channels, a plurality of collection connection channels provided in a corresponding relation with the pressure chambers, connected to the pressure chambers, and configured to collect the liquid respectively from the plurality of pressure chambers, and a plurality of common collection channels connected to the plurality of collection connection channels, the common supply channel and the common collection channel extend in a direction crossing the main scanning direction and the ejection direction, the circulation pump is located inside the liquid ejection unit, and the ejection module is connected to two or more circulation units.

12. The liquid ejection apparatus according to claim 11, wherein the liquid is circulated through a circulation path formed in the liquid ejection unit with the circulation pump.

13. The liquid ejection apparatus according to claim 11, wherein the pressure adjustment unit

comprises a first pressure adjustment unit configured to supply the liquid to the ejection module and a second pressure adjustment unit configured to collect the liquid from the ejection module.
